Q1:JUN 2003 P2

5 A filament lamp operates normally at a potential difference (p.d.) of 6.0 V. The variation with p.d. V of the current I in the lamp is shown in Fig. 5.1.

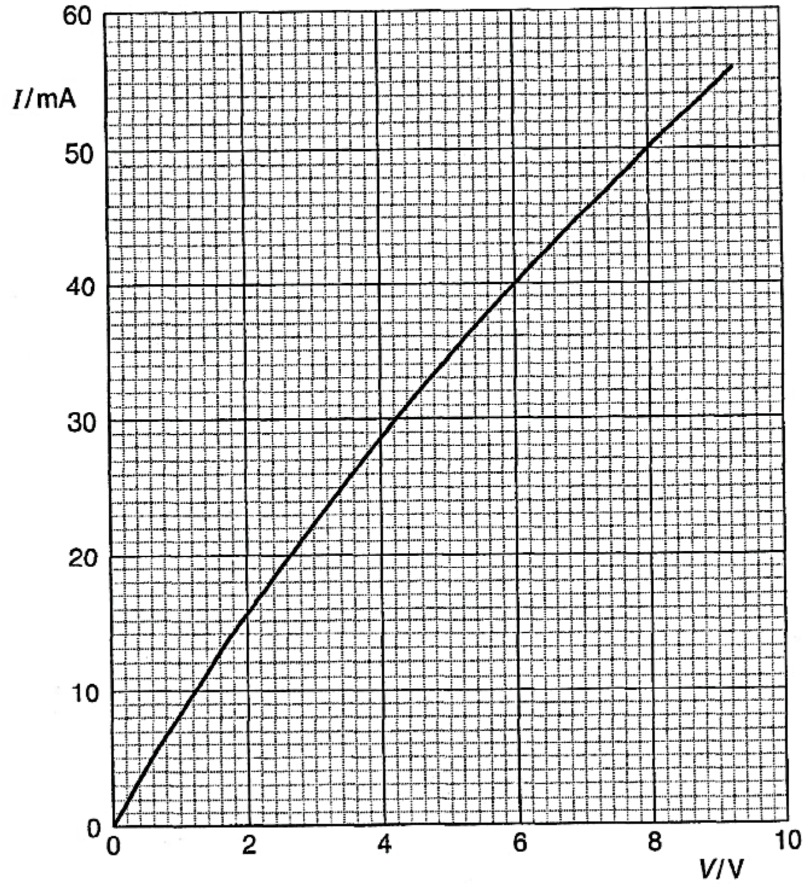

Fig. 5.1

(a) Use Fig. 5.1 to determine, for this lamp,

(i) the resistance when it is operating at a p.d. of 6.0 V,

resistance = ... Ω

(ii) the change in resistance when the p.d. increases from 6.0 V to 8.0 V.

change in resistance = ... Ω
[4]

(b) The lamp is connected into the circuit of Fig. 5.2.

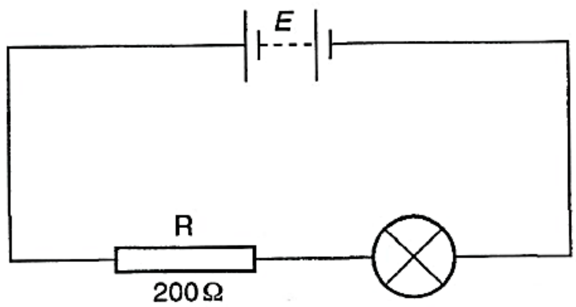

Fig. 5.2

R is a fixed resistor of resistance 200 Ω. The battery has e.m.f. E and negligible internal resistance.

(i) On Fig. 5.1, draw a line to show the variation with p.d. V of the current I in the resistor R.

(ii) Determine the e.m.f. of the battery for the lamp to operate normally.

e.m.f. = ... V
[4]

Q2:NOV 2011 P2

4 **(a)** Describe the movement of electrons when

(i) polythene is rubbed with wool,

____

____

____

(ii) glass is rubbed with silk,

____

____

____ [2]

(b) Describe the electrostatic application in spray painting.

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____ [4]

Q3:NOV 2012 P2

5 Fig.5.1 shows an electrical circuit.

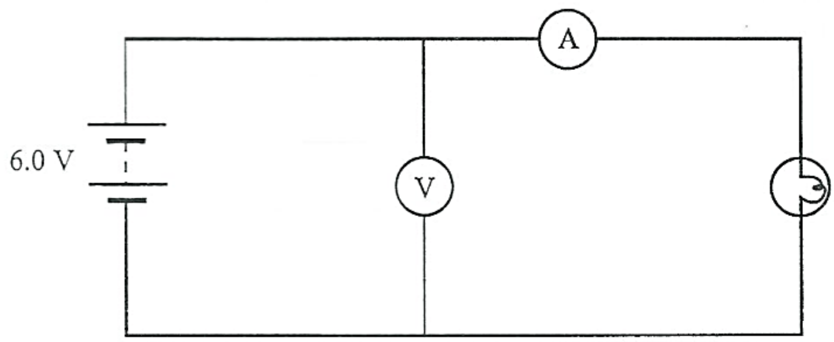

Fig.5.1

The battery in the circuit in **Fig.5.1** has an e.m.f of 6.0 V and drives a current of 0.3 A through the lamp. The voltmeter reading is 4.8 V.

(a) Explain why the voltmeter reading is less than 6.0 V.

____

____

____ [1]

(b) **(i)** Calculate

1. internal resistance, r, of the battery,

r = ____________________

2. energy transfer per second, P, in the lamp.

P = ____________________

(ii) State **two** assumptions you made in order to complete the calculations in **(i)**.

1. ____

____

2. ____

____ [6]

Q4:NOV 2002 P3

4 (a) A body can be charged by friction. With the aid of a diagram, describe another method which can be used to charge an insulated metalic sphere. [2]

(b) (i) Explain how electrostatics is used to extract dust particles from chimneys of factories. [2]

(ii) Define the term *voltage* and state its SI units. [2]

(c) From your definition of voltage in **(b)(ii)** show that electrical power, $P = I^2R$, where R and I of a conductor represent resistance and current respectively. [2]

(d) State **two** differences between voltage and electromotive force(e.m.f). [2]

(e) Write down an equation of the law of conservation of energy for a circuit of e.m.f. E, internal resistance, r and resitors R_1, R_2 and R_3 connected in series. [1]

Q5:NOV 2003 P3

4 (a) State and explain Kirchhoff's laws. [4]

(b) An electric heater, designed to operate on a 240 V supply, consists of a coil of constantan wire of resistivity 4.9×10^{-7} Ωm. The wire used to make the coil is 9.0 m long and has a cross-sectional area of 0.23 mm^2.

Calculate

(i) the resistance of the heating coil,

(ii) the current drawn by the heater,

(iii) the power dissipated as heat in the heater. [6]

Q6:NOV 2012 P5

2 (a) Define *resistance* and *the Ohm*. [2]

(b) Fig 2.1 shows a battery with internal resistance, r, connected to a resistor, R

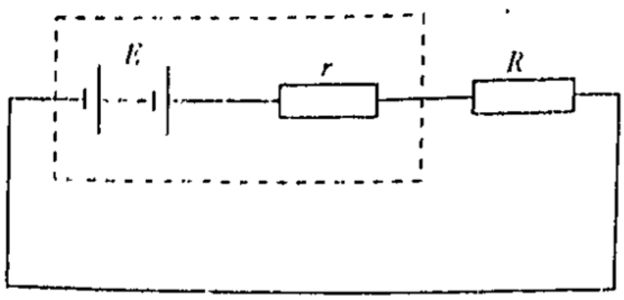

Fig 2.1

Fig. 2.2 shows an incomplete graph that shows the distribution of voltage across the circuit in Fig 2.1.

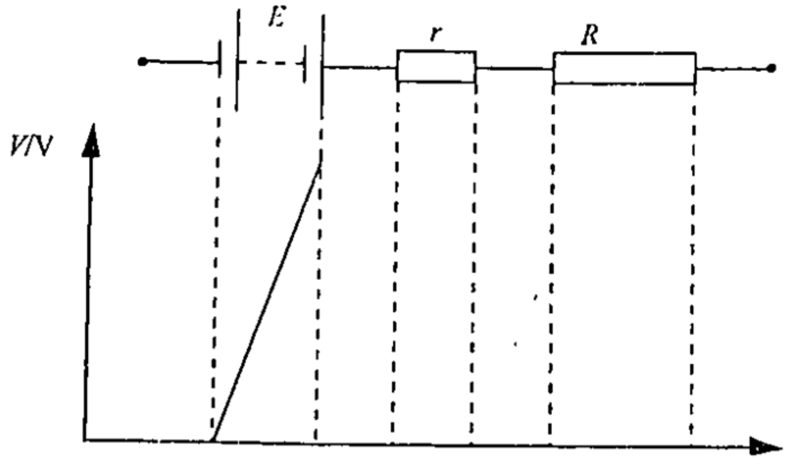

Fig. 2.2

(i) Copy and complete the graph.

(ii) If $E = 12$ V and $r = 0.1\,\Omega$, calculate energy lost per unit charge in the battery itself. [5]

(c) Suggest one advantage and one disadvantage of having a large internal resistance in a battery. [2]

(d) (i) Sketch the I - V characteristics of a filament lamp.

(ii) Explain the shape of your graph. [3]

Q7:JUN 2013 P5

1 (c) (i) State Coulomb's law.

(ii) Compare the electrostatic force and the gravitational force.

(iii) Calculate the force between an electron and a proton 1.0×10^{-10} m apart, in a vacuum.

[5]

Q8:NOV 2015 P5

1 (b) (i) Define *e.m.f.*

(ii) An electric bulb is marked '240 V, 100 W'.

1. Calculate the current in the bulb when operating normally.

2. Describe **three** ways in which thermal energy can be transferred from the bulb filament when operating normally.

[6]

Q8:JUN 2016 P5

1 (a) (i) State **one** practical application of the electrostatic phenomena.

(ii) Describe the construction of a simple lightning conductor.

(iii) Suggest a reason why a human hand can be used to discharge a charged metal rod.

[4]

3 (a) Define *charge* and the *coulomb*. [2]

(b) An electric kettle takes 2.5 minutes to boil 2 litres of water when it is connected to a 240 V supply.

Assume that all the electrical energy is converted into 2.4×10^5 J of heat energy.

(i) Calculate the

1. current drawn from the supply,

2. quantity of electric charge that flowed in the kettle for 2.5 minutes.

(ii) The heating element in the kettle is made of nichrome wire of total length 2.4 m.

Calculate the diameter of the wire used if the resistance of nichrome is 1.1×10^{-6} Ωm

(iii) The diameter of the wire was found to be 0.29 mm using a micrometer screw gauge. This is different from the answer in **b(ii)**.

Give **three** possible reasons for this observation. [10]

Q9:JUN 2018 P5

1 **(a)** **(i)** 1. Sketch the I-V characteristic of a copper wire at constant temperature.

2. Explain the shape of the sketched graph in **(i)** 1.

(ii) **Fig.1.1** shows a thermistor in a circuit.

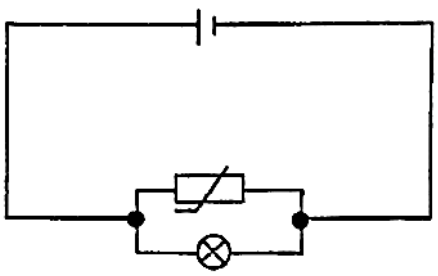

Fig.1.1.

Comment on the brightness of the lamp when the thermistor is immersed in

1. cold water,

2. hot water. [6]

2 **(a)** Distinguish between electrical conductors and insulators using a simple electron model. [2]

(b) **(i)** Explain how charge is conserved during charging by friction.

(ii) Given that an uncharged metal sphere is attached to an insulating stand,

1. describe how the sphere can be charged by induction using a positively charged perspex rod,

2. suggest how the type of charge acquired by the sphere can be verified. [7]

(c) (i) Explain the significance of the sharp points at the top of a lightning conductor.

(ii) Give any **one** merit of electrostatic paint spraying. [3]